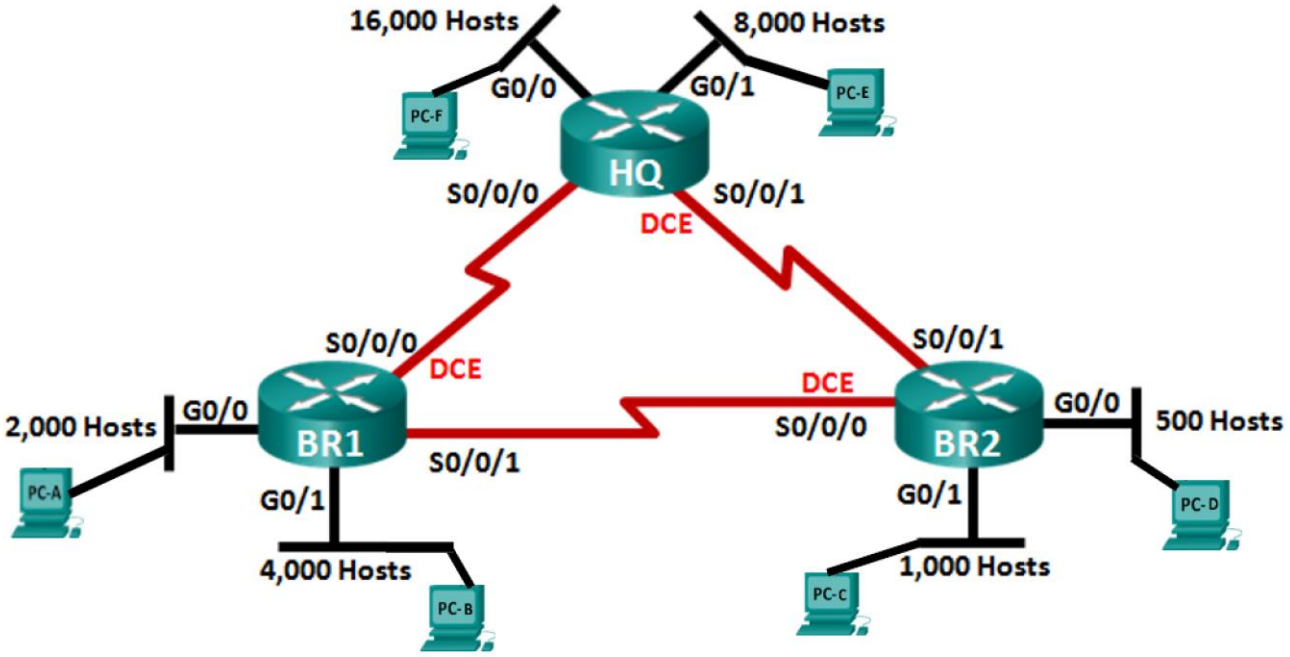
Activity 9.2 – Designing and Implementing a VLSM Addressing Scheme PACKET TRACER

Topology



Objectives

Part 1: Examine Network Requirements

Part 2: Design the VLSM Address Scheme

Part 3: Cable and Configure the IPv4 Network

Background / Scenario

Variable Length Subnet Mask (VLSM) was designed to avoid wasting IP addresses. With VLSM, a network is subnetted and then re-subnetted. This process can be repeated multiple times to create subnets of various sizes based on the number of hosts required in each subnet. Effective use of VLSM requires address planning.

In this lab, use the 172.16.128.0/17 network address to develop an address scheme for the network displayed in the topology diagram. VLSM is used to meet the IPv4 addressing requirements. After you have designed the VLSM address scheme, you will configure the interfaces on the routers with the appropriate IP address information as well as configure the PCs on each network.

Note: Serial cables have two ends, one of which is called DCE and the other is called DTE. The DCE end will provide the clock and other control signals to the DTE end.

Required Resources

1. Packet Tracer 8.2.2

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Part 1: Examine Network Requirements

In Part 1, you will examine the network requirements to develop a VLSM address scheme for the network displayed in the topology diagram using the 172.16.128.0/17 network address.

Step 1: Determine how many host addresses and subnets are available.

How many host addresses are available in a /17 network? _____

What is the total number of host addresses needed in the topology diagram? _____

What is the minimum number of bits needed to support all the hosts? _____

Given that many host-bits, how many bits are left for the network and subnets? _____

What is the subnet mask for a /17 network? _____

What is the binary for the /17 subnet mask? _____

What is the binary for IP address 172.16.128.0? _____

How many subnets are needed in the network topology? _____

Based on the above, you will need to use VLSM to allocate the subnets within the 172.16.128.0/17 network.

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Step 2: Determine the largest subnet.

What is the subnet name of the largest subnet? _____

How many IP addresses are required in the largest subnet? _____

How many bits are required to support that many hosts? _____

What subnet mask can support that many host addresses? _____

How many actual host addresses can that subnet mask support? _____

Can you subnet the 172.16.128.0/17 network address to support this subnet? _____

What are the two network addresses that would result from this subnetting?

Network address 1: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Network address 2: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Use the first network address for this subnet. That leaves the second network for further subnetting.

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Step 3: Determine the second largest subnet.

What is the subnet name of the next largest subnet? _____

How many IP addresses are required in the next largest subnet? _____

How many bits are required to support that many hosts? _____

What subnet mask can support that many host addresses? _____

How many actual host addresses can that subnet mask support? _____

Can you subnet the remaining network address to support this subnet? _____

What are the two network addresses that would result from this subnetting?

Network address 1: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Network address 2: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Use the first network address for this subnet. That leaves the second network for further subnetting.

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Step 4: Determine the next largest subnet.

What is the subnet name of the next largest subnet? _____

How many IP addresses are required in the next largest subnet? _____

How many bits are required to support that many hosts? _____

What subnet mask can support that many host addresses? _____

How many actual host addresses can that subnet mask support? _____

Can you subnet the remaining network address to support this subnet? _____

What are the two network addresses that would result from this subnetting?

Network address 1: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Network address 2: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Use the first network address for this subnet. That leaves the second network for further subnetting.

Step 5: Determine the next largest subnet.

What is the subnet name of the next largest subnet? _____

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How many IP addresses are required in the next largest subnet? _____

How many bits are required to support that many hosts? _____

What subnet mask can support that many host addresses? _____

How many actual host addresses can that subnet mask support? _____

Can you subnet the remaining network address to support this subnet? _____

What are the two network addresses that would result from this subnetting?

Network address 1: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Network address 2: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Use the first network address for this subnet. That leaves the second network for further subnetting.

Step 6: Determine the next largest subnet.

What is the subnet name of the next largest subnet? _____

How many IP addresses are required in the next largest subnet? _____

How many bits are required to support that many hosts? _____

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What subnet mask can support that many host addresses? _____

How many actual host addresses can that subnet mask support? _____

Can you subnet the remaining network address to support this subnet? _____

What are the two network addresses that would result from this subnetting?

Network address 1: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Network address 2: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Use the first network address for this subnet. That leaves the second network for further subnetting.

Step 7: Determine the next largest subnet.

What is the subnet name of the next largest subnet? _____

How many IP addresses are required in the next largest subnet? _____

How many bits are required to support that many hosts? _____

What subnet mask can support that many host addresses? _____

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How many actual host addresses can that subnet mask support? _____

Can you subnet the remaining network address to support this subnet? _____

What are the two network addresses that would result from this subnetting?

Network address 1: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Network address 2: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Use the first network address for this subnet. That leaves the second network for further subnetting.

Step 8: Determine the subnets needed to support the serial links.

We will need to support three router to router networks (serial links). Each of these networks consists of 2 routers (actual hosts). What are the subnet names for the three required remaining router to router networks?

1. _____
2. _____
3. _____

How many IP addresses are required in each of these subnets? _____

How many bits are required to support that many hosts? _____

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What subnet mask can support that many host addresses? _____

How many actual host addresses can that subnet mask support? _____

Can you subnet the remaining network to support these three subnets? _____

Note, by subnetting this way there will be more than enough available subnets to meet the requirement of 3 subnets for the router to router connections.

What are the first four network addresses that would result from this subnetting?

Network address 1: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Network address 2: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Network address 3: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

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Decimal:

Network address 4: Decimal:

Binary:

Subnet mask:

First host address:

Decimal:

Last host address:

Decimal:

Broadcast address:

Decimal:

Use the first network address for HQ-S0/0/0_To_BR1_S/0/0/0.

Use the second network address for BR2-S0/0/1_To_HQ-S0/0/1.

Use the third network address for BR1-S0/0/1_To_BR2-S/0/0/0.

The remaining (fourth) network address is unused for now.

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Part 2: Design the VLSM Address Scheme

Step 1: Calculate the subnet information.

Subnet Description	Number of Hosts Needed	Network Address	Prefix length	First Host Address	Broadcast Address
HQ G0/0	16,000				
HQ G0/1	8,000				
BR1 G0/1	4,000				
BR1 G0/0	2,000				
BR2 G0/1	1,000				
BR2 G0/0	500				
HQ S0/0/0 – BR1 S0/0/0	2				
HQ S0/0/1 – BR2 S0/0/1	2				
BR1 S0/0/1 – BR2 S0/0/0	2				

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Step 2: Complete the device interface address table.

Serial interfaces:

HQ-BR1 network:

HQ must be given the first available host address on the Serial link to BR1. BR1 must be given the second available host address on the Serial link to HQ.

HQ-BR2 network:

HQ must be given the first available host address on the Serial link to BR2. BR2 must be given the second available host address on the Serial link to HQ.

BR1-BR2 network:

BR1 must be given the first available host address for the serial link to BR2. BR2 must be given the second available host address for the serial link to BR1.

Ethernet interfaces:

Each G0/0 and G0/1 interface will get the first available address in the network that connects to them.

PCs:

Each PC will get the 5th available host address on their network.

Device	Interface	IP Address	Subnet Mask	Default gateway	Connects to
HQ	G0/0			N/A	16,000 Host LAN
	G0/1			N/A	8,000 Host LAN
	S0/0/0			N/A	BR1 S0/0/0
	S0/0/1			N/A	BR2 S0/0/1
BR1	G0/0			N/A	2,000 Host LAN
	G0/1			N/A	4,000 Host LAN
	S0/0/0			N/A	HQ S0/0/0
	S0/0/1			N/A	BR2 S0/0/0
BR2	G0/0			N/A	500 Host LAN
	G0/1			N/A	1,000 Host LAN
	S0/0/0			N/A	BR1 S0/0/1
	S0/0/1			N/A	HQ S0/0/1
PC-A	NIC				BR1 G0/0
PC-B	NIC				BR1 G0/1
PC-C	NIC				BR2 G0/1
PC-D	NIC				BR2 G0/0
PC-E	NIC				HQ G0/1
PC-F	NIC				HQ G0/0

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Part 3: Cable and Configure the IPv4 Network

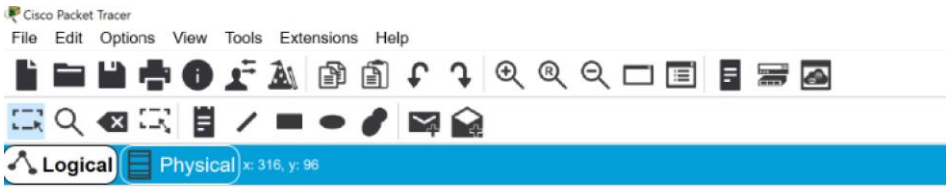
In Part 3, you will cable the network topology and configure the three routers using the VLSM address scheme that you developed in Part 2.

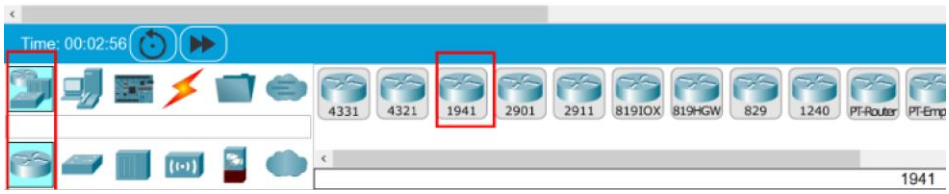
Installing the serial ports into the routers:

IMPORTANT: Install serial ports into each router (see associated video for this section). If you do not complete this step you will not be able to create the serial connections between the routers.

The default Packet Tracer 1941 router does not have the serial port module installed. Before you can use serial ports on a router you will need to install the module first. The following steps illustrate the process:

In Packet Tracer, create a 1941 router:





Note: Packet Tracer attempts to provide a realistic real-world simulation of devices. This means that you can update the devices by installing specific networking modules. However, it also means that before you do the installation you must prepare the device so the installation can be done safely. In this case, it means that the router must be powered off before installing the serial ports. The next couple of steps go through the installation process. The steps that we will follow will be:

1. Turn off the power on the router
2. Install the Serial port module
3. Turn on the power

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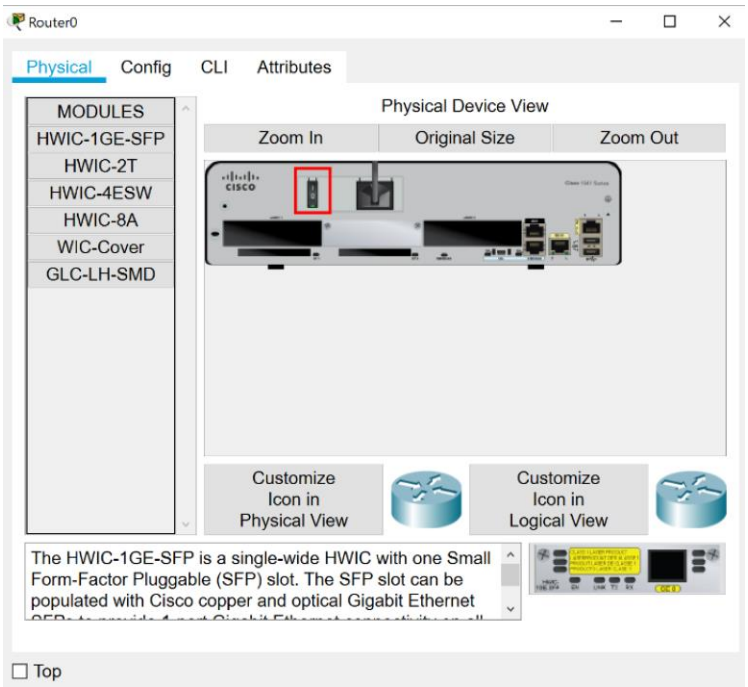
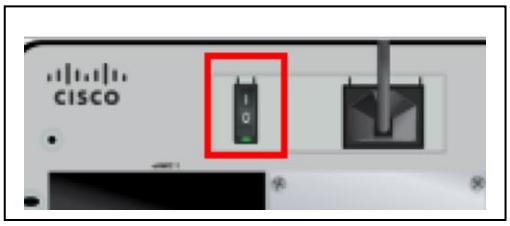
Turn off the power on the router

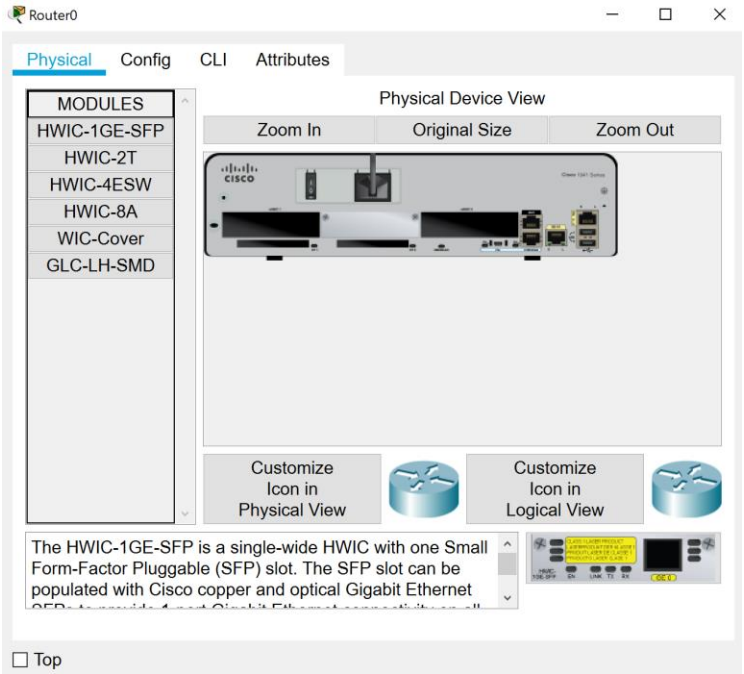
Left click on the router to open the Router properties menu. In the Physical menu, you will see a picture of the back of the router. On the top left there is a power button. When you create a router in Packet Tracer, the power button will be set to on (small green dot at the bottom of the power button).

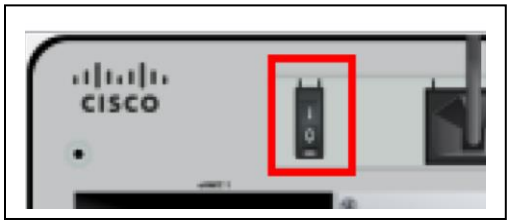
Note: You can use the “Zoom in” button above the router image if the image is too small to see clearly.





Left-click on the power button to turn off the router (the small green dot will disappear). The power is now off.





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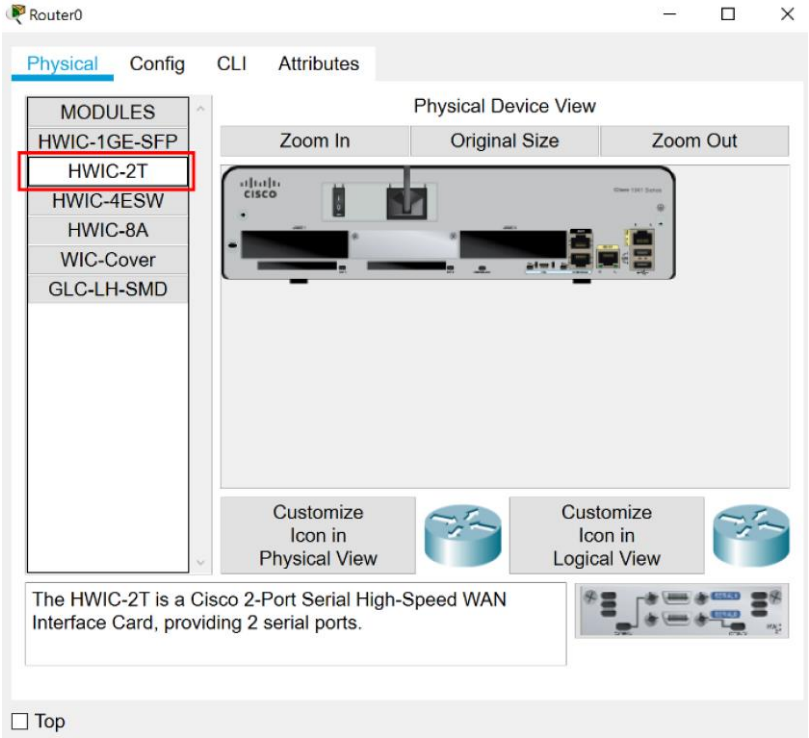
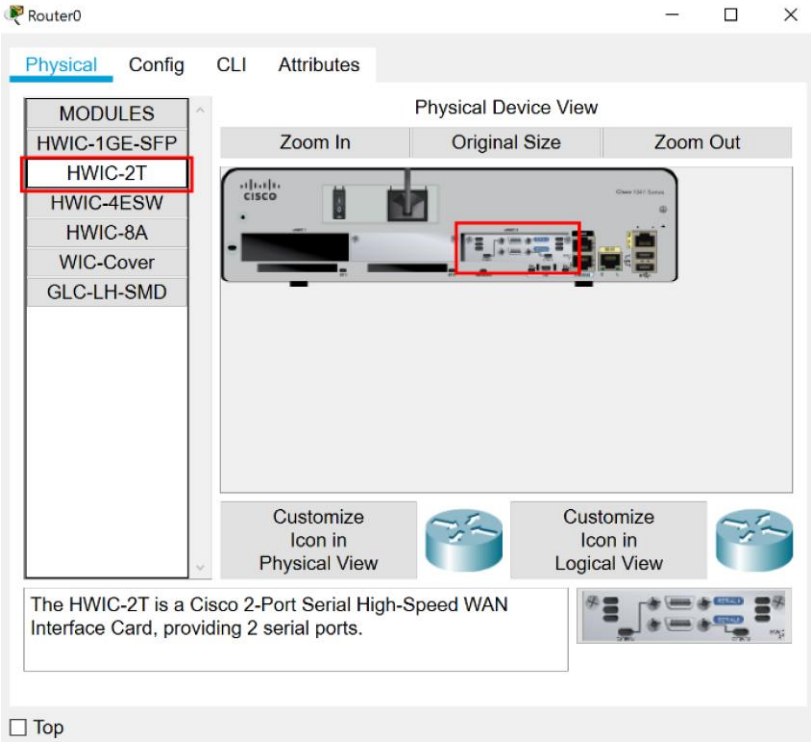
Insert the Serial port module (HWIC-2T)

The next step is to add the Serial port module to the router. Left-Click on the HWIC-2T item to the left of the router image:



Drag and drop the HWIC-2T box onto the open slot (black box) on the bottom right of the router image:



The Serial ports S0/0/0 and S0/0/1 have now been installed on the router.

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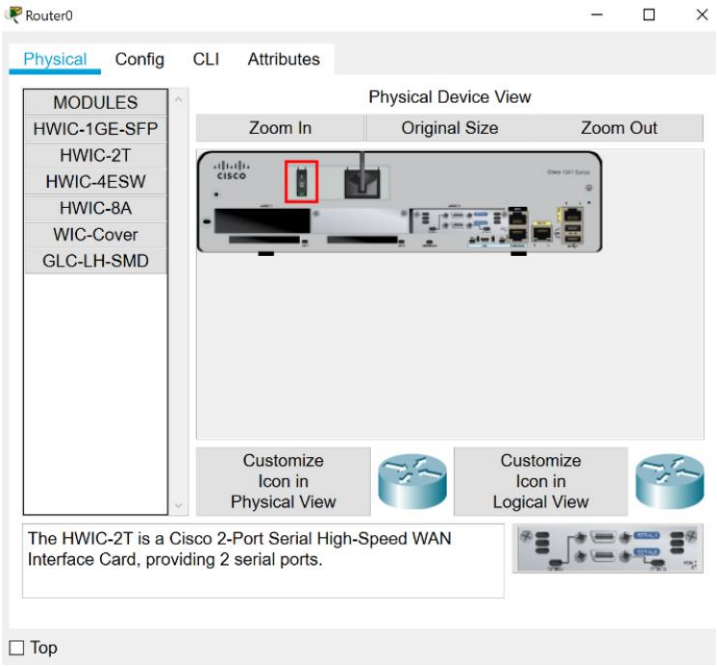
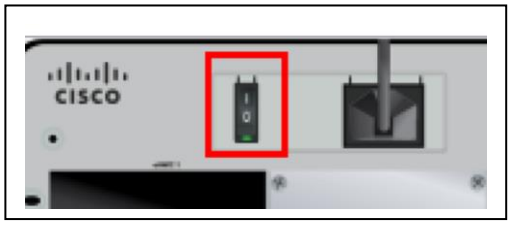
Turn the power on to the router

The final step for installing the Serial ports is to turn the power back on to the router. Left click on the power button to turn on the router. You should see the green dot at the bottom of the power button when the router is on.



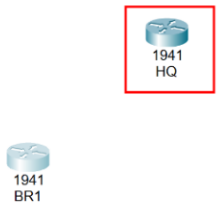


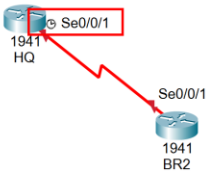
You have now installed the Serial ports onto this router. Repeat this process for each other router in the network.

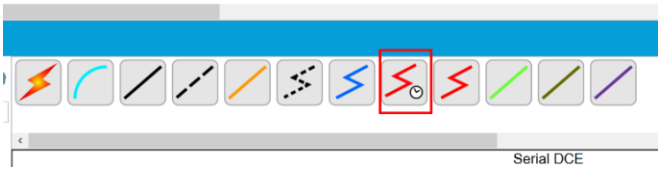
Cable the serial cables between the routers:

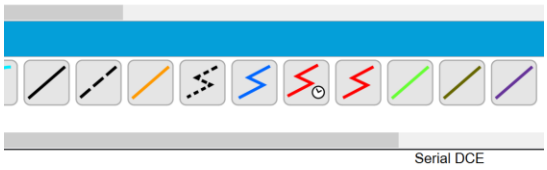
Serial cables have two ends, one of which is called DCE and the other is called DTE. The DCE end will provide the clock and other control signals to the DTE end.

In Packet Tracer use the DCE cable connect the HQ router S0/0/1 to the BR2 router S0/0/1 (start the cable from the HQ router). You should see a little clock icon next to S0/0/1 on the HQ router indicating that is the DCE end.









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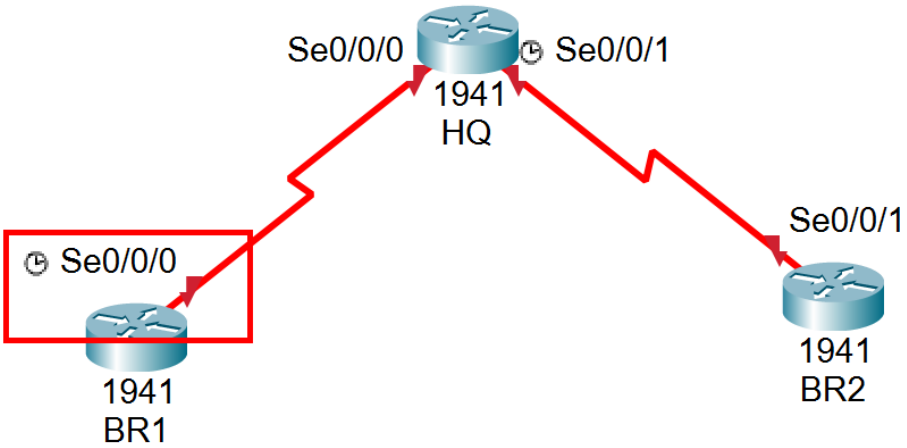
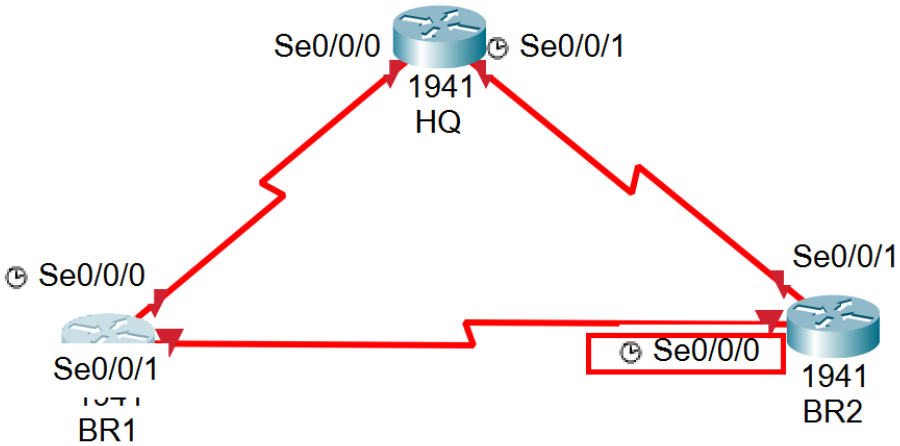
Now put a DCE cable from BR1 S0/0/0 to HQ S0/0/0 (start the cable from BR1) and make sure that the clock icon is next to BR1 S0/0/0.



Now put a DCE cable from BR2 S0/0/0 to BR1 S0/0/1 (start the cable from BR2) and make sure that the clock icon is next to BR2 S0/0/0.



Note: For this lab we will use the default clock rate of 2000000 but you could modify this on the DCE end of each serial cable based on your network requirements. For your reference, the command to change the clock rate is:

```
HQ(config-if)# clock rate [rate]
```

Do not change the clock rate for this lab.

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Step 2: Configure the routers.

Fully configure the routers as per the standard configuration steps discussed in previous labs and save the configuration

a. Configure the routers:

   Configure the routers as per the standard configuration in previous labs:

   1) Assign a device name to the router.
   2) Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.
   3) Assign **class** as the privileged EXEC encrypted password.
   4) Assign **cisco** as the console password and enable login.
   5) Assign **cisco** as the VTY password and enable login . Verify the number of vty lines are on the router.
   6) Encrypt the clear text passwords.
   7) Create a banner that warns anyone accessing the device:

      Unauthorised access is prohibited and you will be prosecuted.

   8) Assign *IPv4* addresses to all interfaces on Router as per the addressing table and enable them.

      Configure the serial interfaces the same way that you would configure the other router interfaces:

      Example:

```
HQ(config)# interface S0/0/0
R1(config-if)# ip address [ip address] [subnet mask]
R1(config-if)# no shutdown
R1(config-if)# exit
```

      Note: Both ends of the Serial interfaces must be configured before the interface will show as up/up when viewing with the ip interface brief command. When you enter the ip address and no shutdown command on the first end the interface will stay down and you will likely get a similar message to this:

```
BR1(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
BR1(config-if)#
```

      After configuring the second end of the serial cable you should see a similar message to this:

```
BR2(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0,
changed state to up
```

      it can also sometimes take 20 seconds or so for the connection to be tested and be established.

      Note2: If you are finding that a connection does not appear to be updating, you can try saving the file, closing it and reopening the file to try to force the simulation to update.

Step 3: Configure the PC interfaces.

a. Configure the IP address, subnet mask, and default gateway settings on PC-A through PC-F

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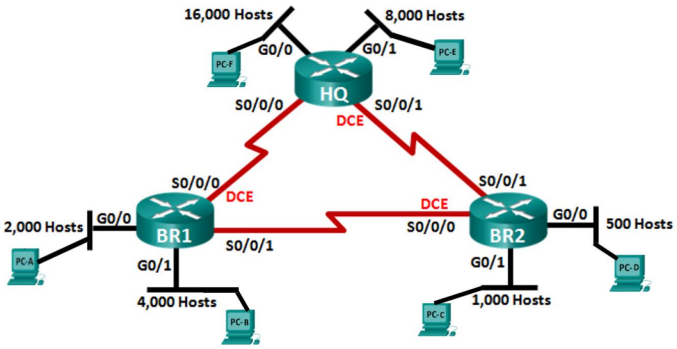
Part 4: Test and Troubleshoot the Network

Note: For this exercise, the routers do not have additional routing information configured for the remote networks. You will therefore not be able to ping PCs or interfaces that are not on your network and not attached directly to your router. Configuring routing information for remote networks is outside the scope of this course.



For example, from PC-A you can ping to the following destinations with only those directly connected to the same router (BR1) successfully:

Source	Destination	Success	Reason
PC-A	BR1 G0/0	Yes	Default gateway for PC-A
PC-A	BR1 G0/1	Yes	both source and destination directly connected to PC-A router BR1
PC-A	BR1 S0/0/0	Yes	both source and destination directly connected to PC-A router BR1
PC-A	BR1 S0/0/1	Yes	Both source and destination directly connected to PC-A router BR1
PC-A	BR2 G0/0	No	Destination on different router without additional routing information
PC-A	BR2 G0/1	No	Destination on different router without additional routing information
PC-A	BR2 S0/0/0	No	Destination on different router without additional routing information
PC-A	BR2 S0/0/1	No	Destination on different router without additional routing information
PC-A	HQ G0/0	No	Destination on different router without additional routing information
PC-A	HQ G0/1	No	Destination on different router without additional routing information
PC-A	HQ S0/0/0	No	Destination on different router without additional routing information
PC-A	HQ S0/0/1	No	Destination on different router without additional routing information
PC-A	PC-B	Yes	both networks directly connected to PC-A router BR1
PC-A	PC-C	No	Destination on different router without additional routing information
PC-A	PC-D	No	Destination on different router without additional routing information
PC-A	PC-E	No	Destination on different router without additional routing information
PC-A	PC-F	No	Destination on different router without additional routing information

Verify connectivity for the devices that are on networks that are directly attached to the same router.

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